



Customer Shopping Behavior Analysis

This project analyzes transactional data from 3,900 purchases across multiple product categories to uncover spending patterns, customer segments, product preferences, and subscription behavior. Findings support strategic decisions for marketing, merchandising, pricing, and retention.

Dataset Summary

Dataset: 3,900 rows × 18 columns. Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color), and behavior signals (discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type). Review Rating contains 37 missing values.





Data Preparation & Cleaning (Python)

Loaded data with pandas and profiled structure via `df.info()` and `df.describe()`. Handled missing Review Rating values by imputing the median rating per product category. Standardized column names to snake_case for readability. Engineered age_group bins and purchase_frequency_days. Determined promo_code_used redundant with discount_applied and dropped promo_code_used. Connected cleaned DataFrame to PostgreSQL for downstream SQL analysis.

Exploratory Summary Statistics

Key numeric summaries: mean age ~44.07 (std 15.21), mean purchase amount \$59.76 (std \$23.69), review rating mean 3.75 (std 0.72). Categorical highlights: Male was the most frequent gender (2,652), most common item Blouse (171), top location example Montana (96). There are 6 payment methods and a dominant shipping type of Free Shipping (2,847 occurrences).

Rows & Columns

3,900 rows × 18 columns

Missing Values

37 missing in Review Rating

Avg Purchase

\$59.76 (std \$23.69)

SQL Analysis — Revenue & Demographics

Revenue by gender: Female total revenue 75,191; Male total revenue 157,890. Revenue by age group: Young Adult 62,143; Middle-aged 59,197; Adult 55,978; Senior 55,763. Subscribers: 1,053 customers averaging \$59.49 (total revenue 62,645); Non-subscribers: 2,847 customers averaging \$59.87 (total revenue 170,436).



Product & Rating Insights

Top 5 products by average rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78). Top products by orders (category-level examples): Accessories — Jewelry (171), Sunglasses (161), Belt (161). Clothing — Blouse (171), Pants (171), Shirt (169). Footwear — Sandals (160), Shoes (150), Sneakers (145).



Gloves

Highest avg rating 3.86



Sandals

Avg rating 3.84



Boots

Avg rating 3.82

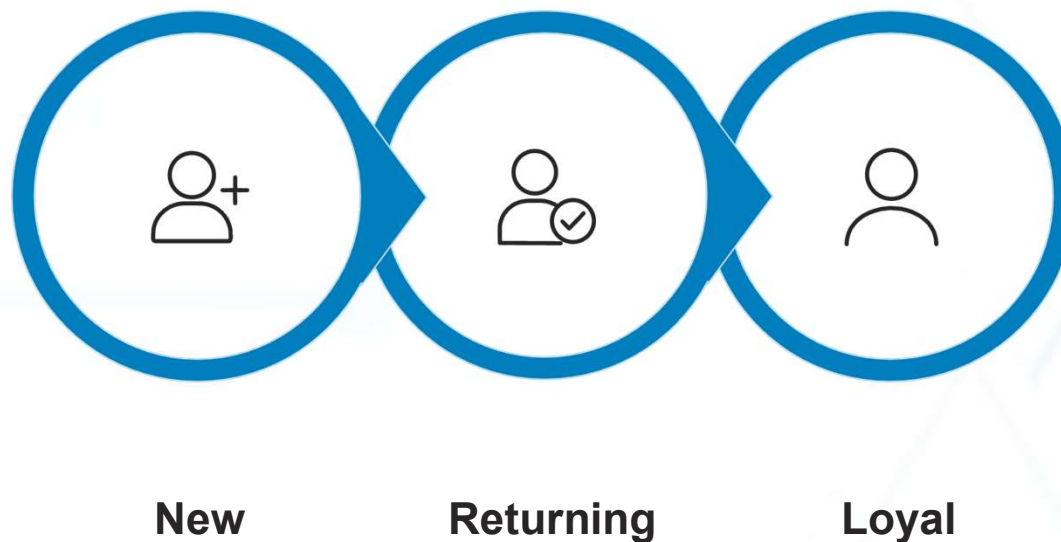
Discounts, Shipping & Purchase Behavior

Discount-dependent products (highest discount rates): Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%. Shipping comparison shows similar average purchase amounts: Standard \$58.46 vs Express \$60.48. Identified 839 rows of discount-users who spent above average in the query output. Repeat buyer counts: Loyal 3,116; Returning 701; New 83.



Customer Segmentation & Subscriptions

Segment counts: Loyal 3,116; Returning 701; New 83. Subscription penetration: 27% Yes / 73% No (donut). Repeat buyers (>5 purchases) are more prevalent in loyalty groups: among repeat buyers subscription status shows higher subscription proportions. Subscribers show slightly lower avg spend than non-subscribers but represent meaningful recurring relationships.



The diagram highlights progression from New to Loyal customers and rising propensity to subscribe and repeat purchase behavior.

Power BI Dashboard Highlights

Interactive dashboard visualized: total customers 3.9K, average purchase \$59.76, average review rating 3.75. Visuals include subscription donut (Yes 27% / No 73%), revenue and sales by category (Clothing highest revenue ~90K; Clothing highest sales ~1,800), and revenue/sales by age group (Young Adult highest). Dashboard supports filter by subscription, gender, category, and shipping type.



Business Recommendations & Next Steps

Boost Subscriptions

Promote exclusive subscriber benefits to grow recurring revenue and engagement.

Customer Loyalty Programs

Reward repeat buyers to convert Returning into Loyal segments and increase lifetime value.

Review Discount Policy

Balance discount-driven sales (notably for Hat, Sneakers, Coat) with margin controls and targeted promotions.

Product Positioning

Feature top-rated and top-selling products in campaigns to capitalize on positive reviews and demand.

Targeted Marketing

Focus campaigns on high-revenue age groups and express-shipping users to optimize acquisition and conversion.